Data Mining 1 Project

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1 DATA UNDERSTANDING AND PREPARATION

The goal of the project is to analyze the IMDb Dataset, which contains data about movies, TV shows, and other forms of visual entertainment, along with their ratings. Each record includes key information about the title, as well as insights into critical aspects such as awards and reviews, as well as statistical ratings and other metadata. The dataset is updated as of September 1, 2024.

1.1 Data Semantics and First Global Exploration of the Dataset

The IMDb Dataset contains 16431 records and 23 attributes, of which 10 are categorical and 13 numeric. They are listed in Tables 1 and 2. Among the **numerical attributes**, most of them are discrete and ratio-scaled, as they represent counts, with *startYear* and *endYear* being interval values. Only *runtimeMinutes* can be classified as a continuous attribute, as it represents a time duration, even though it only assumes integer values in the dataset. Among the **categorical attributes**, *canHaveEpisodes*, *isRatable* and *isAdult* are **binary**, *rating* , *worstRating* and *bestRating* are **ordinal**, while the rest are all **nominal**.

Looking at the records' data types, we can observe some syntactic inaccuracies: *isAdult* should be converted from integer to boolean, whereas *rating* assumes ten categorical values of the form *'(0, 1]', '(1, 2]', ..., '(9, 10]'*. We can assume they represent the binning of a continuous rating, therefore we map them into an integer space [1,10] to ease later analysis. Just by looking at the head of the dataframe, we can also tell that *countryOfOrigin* and *genres* contain collections of categorical values; therefore, we transform them into lists to allow for easier handling. Some discrete numerical attributes (*endYear*, *awardWins*) are stored as float rather than boolean because `int64` does not support NaN values; however, we do not deem it necessary to convert them.

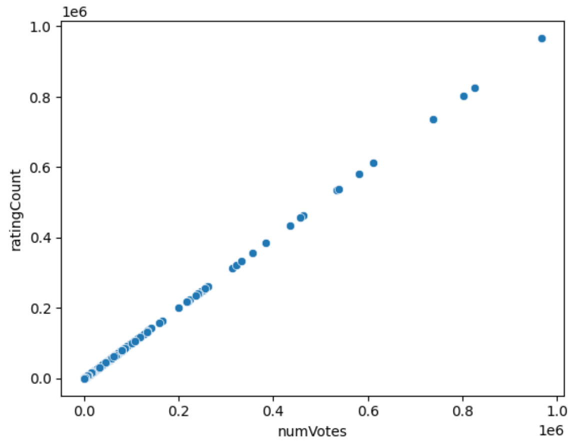

Figure 1: Scatterplot ratingCount~numVotes.

A first global analysis of the dataframe's summary and descriptive statistics already gives us valuable insight into which variables have missing values[1], as reported in Table 3, and into which features are uninformative: *bestRating*, *worstRating*, and *isRatable* can be safely dropped, as they only assume one value for all records (10, 1, and 1, respectively). We can also see that *numVotes* and *ratingCount* share very similar, when not identical, statistics. The scatterplot in Figure 1 proves their correlation close to 1, which leads us to drop *ratingCount* and only keep *numVotes*.

1.2 Distribution of categorical variables and statistics

In an effort to analyze the univariate distribution of categorical variables, we produced barplots for *titleType* (Figure 2), *countryOfOrigin* (Figure 3), and *genres* (Figure 4). All distributions are imbalanced, being dominated by few very common classes: *movie* and *tvepisode*, the *US*, *drama* and *comedy*, respectively. A barplot for *rating*[2], in Figure 5,

1 As the missing values were recorded inconsistently in the dataset, we used the `read_csv` function, passing a list of strings to recognize as missing values to the parameter `na_values`.

2 In this data exploration phase, we analyzed this ordinal variable both from a categorical and a numeric point of view, as to maximize insights.

Feature	Description	Type	Pandas dtype
runtimeMinutes	Primary runtime of the title, in minutes.	Continuous. Ratio-scaled.	float64
startYear	Represents the release year of a title. In the case of TV Series, it is the series' start year.	Discrete. Interval.	int64
endYear	TV Series end year.	Discrete. Interval.	float64
ratingCount	The total number of user ratings submitted for the title.	Discrete. Ratio-scaled.	int64
numVotes	Number of votes the title has received.	Discrete. Ratio-scaled.	int64
numRegions	The regions number for this version of the title.	Discrete. Ratio-scaled.	int64
totalImages	Total Number of Images for the title within the IMDb title page.	Discrete. Ratio-scaled.	int64
totalVideos	Total Number of Videos for the title within the IMDb title page.	Discrete. Ratio-scaled.	int64
totalCredits	Total Number of Credits for the title.	Discrete. Ratio-scaled.	int64
criticsReviewTotal	Total Number of Critic Reviews.	Discrete. Ratio-scaled.	int64
awardWins	Number of awards the title won.	Discrete. Ratio-scaled.	float64
awardNominationExclWins	Number of award nominations excluding wins.	Discrete. Ratio-scaled.	int64
userReviewsTotal	Total Number of Users Reviews.	Discrete. Ratio-scaled.	int64

Table 1: Numeric attributes.

Feature	Description	Type	Pandas dtype
originalTitle	Original title, in the original language.	Categorical; mostly unique classes (i.e., names).	object
titleType	The type/format of the title (e.g., movie, short, episode, video, etc.).	Categorical, 10 classes.	object
countryOfOrigin	The country where the title was primarily produced. Some titles can belong to more than one class.	Categorical, 153 classes. A record can belong to more than one class.	object
genres	The genre(s) associated with the title (e.g., drama, comedy, action). Some titles can belong to more than one class.	Categorical, 29 classes. A record can belong to more than one class.	object
canHaveEpisodes	Whether or not the title can have episodes.	Asymmetric binary.	bool
isRatable	Whether or not the title can be rated by users.	Asymmetric binary.	bool
isAdult	Whether or not the title is for adults.	Asymmetric binary.	int64
rating	IMDB title rating class.	Ordinal, 10 values.	object
worstRating	Worst title rating.	Ordinal, supposedly [1,10].	int64
bestRating	Best title rating.	Ordinal, supposedly [1,10].	int64

Table 2: Categorical, ordinal, and binary attributes.

Table 3: Features with missing values.

Feature	Non-Null Records
endYear	814
runtimeMinutes	11579
awardWins	13813
genres	16049

shows a left-skewed distribution, with the mode being the [7,8) category.

By looking at the categories for the different variables, we question the informativeness of *canHaveEpisodes* and *isAdult*. The first assumes positive values only for the title types *tvSeries* and *tvMiniSeries*: we keep it, as an explicit feature of serialized content. *isAdult* proves to be redun-

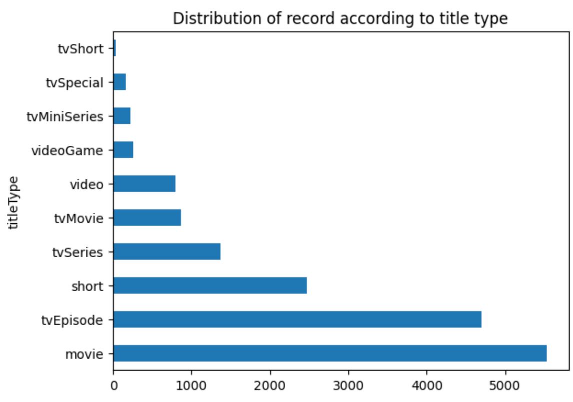

Figure 2: Barplot for *titleType*.

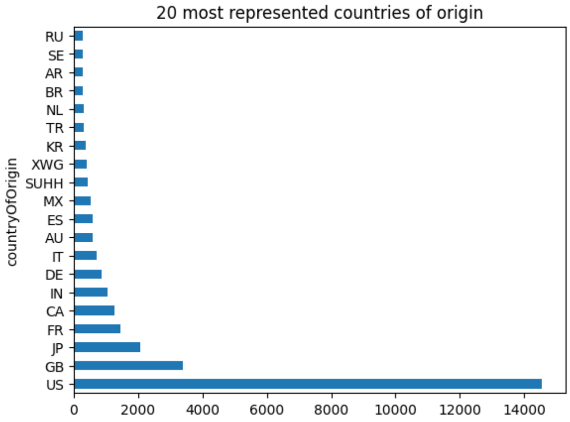

Figure 3: Barplot for *countryOfOrigin* (top 20).

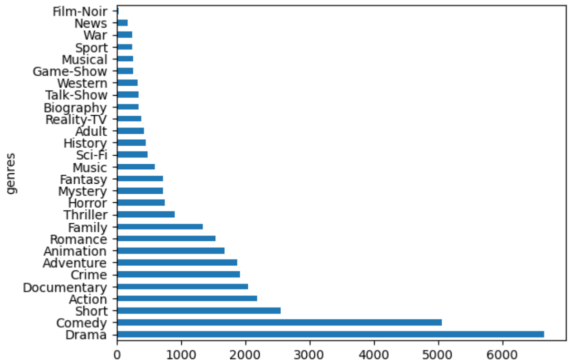

Figure 4: Barplot for *genres*.

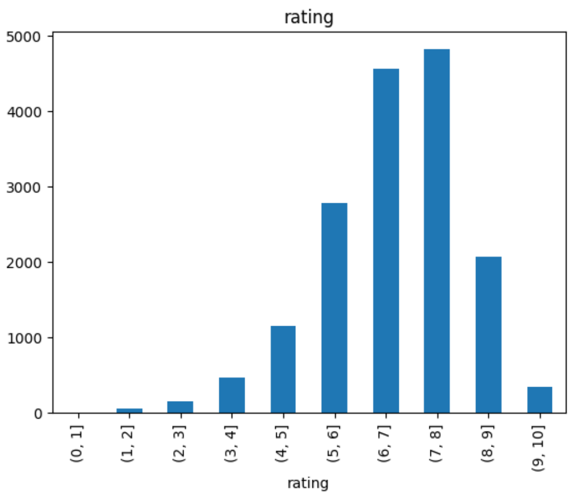

Figure 5: Barplot for *rating*.

dant as its information is conveyed by the *adult* category in *genres*, therefore we drop it.

1.3 Distribution of numeric variables and statistics; transformation of variables

Before analyzing numeric variables, we add the following features:

- *moreCountriesOfOrigin*: the number of countries listed for the record for the variable *countryOfOrigin*. For the training set, the values range from 1 to 10.
- *numGenres*: the number of genres listed in *genres*. For the training set, the values range from 1 to 3.
- *reviewsTotal*: the sum of *criticReviewsTotal* and *userReviewsTotal*.
- *criticReviewsRatio*: the proportion of review from critics (*criticsReviewTotal*) on the total number of reviews of a record (*reviewsTotal*). For unreviewed records, it is set to zero.
- *awardsAndNominations*: the sum of *awardWins* and *awardNominationsExcludeWins*.

Moreover, we use the available interval features, *startYear* and *endYear* to generate the timeline graph in Figure 6. We can observe how the number of titles being released each year has consistently grown until the end of the 2010s, with two noticeable dips in this trend: around 2001 (which could be linked to the aftermath of 9/11) and after 2020 (probably due to the Covid-19 pandemic). The time series for *endYear* follows the same trend on a smaller scale: we suspect a positive correlation between the two variables. There are no end dates available before the 1940s, which is understandable, as serialized content was not produced until around the 1930s.

In order to have a first global view of both the univariate distribution of the numeric variables singularly, and of possible interesting pairwise correlations, we build a pairplot with a KDE graph for each variable in the diagonal. Our previous intuition about a positive correlation between *startYear* and *endYear* is proved by the scatterplot in Figure 7, therefore we drop *endYear*, which is more problematic due to the large number of missing values.

Most numeric features (*awardWins*, *numVotes*, *totalImages*, *totalVideos*, *totalCredits*, *criticReviewsTotal*, *awardNominationsExcludeWins*, *awardsAndNominations*, *numRegions*, *userReviewsTotal*, *num-*

CountriesOfOrigin, *reviewsTotal*) have heavily right-skewed distributions. It makes sense that the distribution of these types of features would approximate a power law. We can address this problem using a log transformation of these features.

On the other hand, *runtimeMinutes* also has a right-skewed distribution, but it really doesn't make sense for it to approximate a power law. When excluding outliers (values higher than 3rd quartile + 1.5 IQR), we can see that, for the most part, runtimes are dependent on the title type and that the distribution of the runtime for each title type approximates a bell curve (Figure 8); therefore, we prefer not to log transform this variable and be mindful of the presence of outliers. Examining the titles with a longer runtime reveals that part of the outliers is due to the inconsistent strategy used for the computation of the runtime for serialized titles (series and miniseries): in some cases, the episode runtime is provided, in others, it reports the runtime for the entire series, i.e. the sum of all its episodes' runtimes (some examples are reported in Table 4).

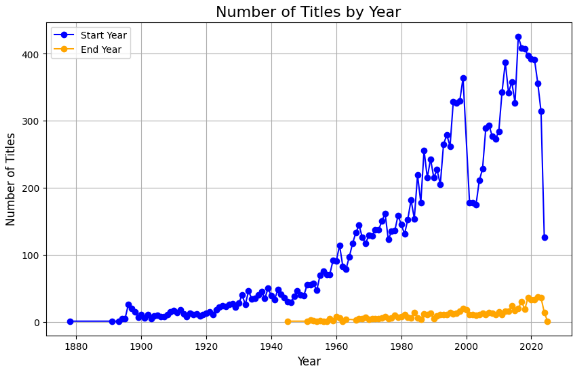

Figure 6: Time series: number of titles being released and ended by year.

originalTitle	**runtimeMinutes**	**titleType**
Alim Day1	3000	tvSeries
Jerry Lewis MDA Labor Day Telethon	1290	tvSeries
Voice of the Planet	600	tvMiniSeries
Orbius	570	movie
Heritage: Civilization and the Jews	540	tvSeries

Table 4: Top 5 titles with the longest runtime.

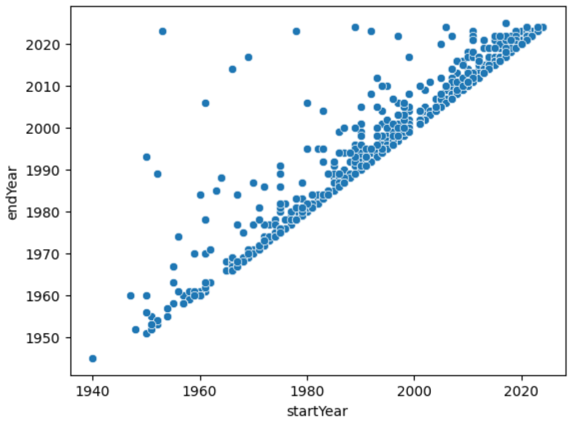

Figure 7: Scatterplot endYear~startYear.

1.4 Handling Missing Values

As explained in Subsection 1.1 and summarized in Table 3, four variables in the dataset have been found to contain missing values: *endYear*, *runtimeMinutes*, *awardWins*, *genres*.

In Subsection 1.3, we opted to drop *endYear* because its value was missing for most records. This decision is corroborated by the fact that, on one side, it heavily correlates with *startYear*, and on the other, the only records that do have an end year are of the type *tvSeries* and *tvMiniSeries*, and the information about the serial nature of media is already carried by the feature *canHaveEpisodes*.

awardWins has almost three thousand missing values. Given its highly unbalanced distribution, with the large majority of the records having no awards, it makes sense to fill them with the most frequent value, i.e. zero.

The stratified barplot in Figure 9 shows that there is some variation in the frequency of differ-

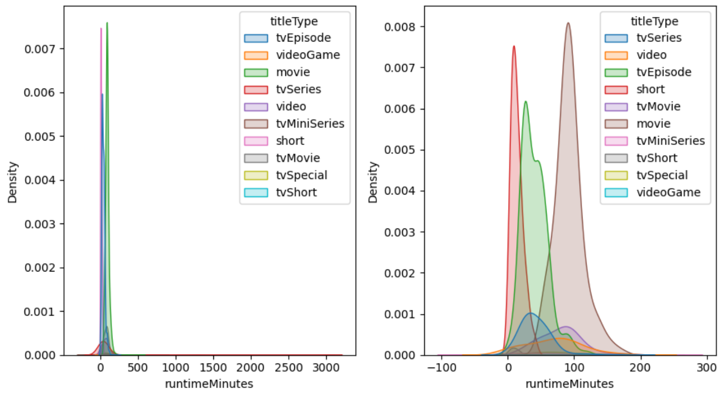

Figure 8: KDE for runtimeMinutes per title type, including and excluding outliers.

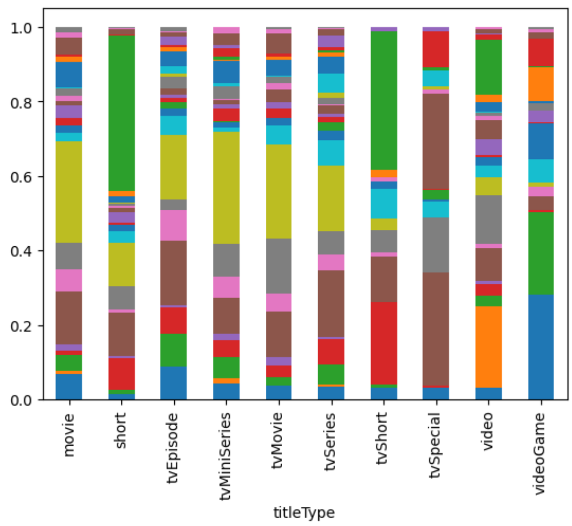

Figure 9: Title type frequency grouped by genre.

ent genres with respect to *titleType*. We decided to fill the 382 missing values for *genres* with the most frequent list of genres given the title type of the record, as shown in Table 5.

titleType	top	# NaN
movie	[Drama]	230
tvSeries	[Comedy]	53
tvMovie	[Drama]	25
video	[Adult]	23
tvSpecial	[Comedy]	18
tvMiniSeries	[Drama]	15
videoGame	[Action, Adventure, Fantasy]	12
tvEpisode	[Comedy]	6

Table 5: Top genre for each title type with missing values for *genres*.

As previously mentioned in Subsection 1.3, *runtimeMinutes* is dependent on the *titleType* of a record (cf. Figure 8): movies are generally longer than TV episodes, and shorts are typically shorter than both. Since we have some missing values for *runtimeMinutes*, we can use the median value for the respective type to fill them (Table 6).

titleType	median runtime
movie	90
short	12
tvEpisode	40
tvMiniSeries	60
tvMovie	86
tvSeries	31
tvShort	10
tvSpecial	60
video	76
videoGame	28

Table 6: Median runtime (in minutes) by title type.

1.5 Pairwise correlations and eventual elimination of variables

We observed pairwise linear correlation among numerical variables (after operating the log transformation detailed in Subsection 1.3) by computing their correlation heatmap, displaying Pearson's correlation coefficient for each pair of features (Figure 10).

numVotes, *userReviewsTotal*, *criticReviewsTotal* and *reviewsTotal* are all highly positively correlated with each other (with a correlation coefficient greater than 70). We only keep *numVotes*, which has the least skewed distribution.

Since the distribution of awards and nominations is not simply skewed, but the majority of records received neither, we binarize *awardsAndNominations* (Table 7) and drop both *awardWins* and *awardNominationsExcludeWins*.

Again, the majority of records do not have videos: we drop *totalVideos* and substitute it with a boolean feature, *hasVideos* (Table 8).

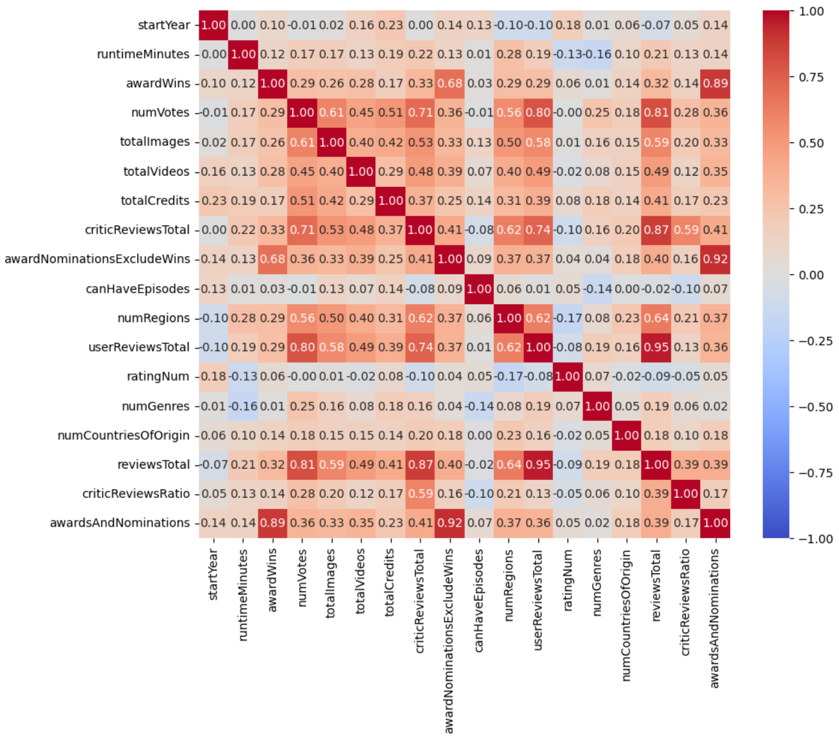

Figure 10: Correlation heatmap before attribute removal.

awardsAndNominations	
False	13692
True	2739

Table 7: *awardsAndNominations* boolean values counts.

hasVideos	
False	14821
True	1610

Table 8: *hasVideos* boolean values counts.

Similarly, most records only have one country of origin, thus we substitute *numCountriesOfOrigin* with the boolean *moreCountriesOfOrigin* (Table 9).

moreCountriesOfOrigin	
False	15285
True	1146

Table 9: *moreCountriesOfOrigin* boolean values counts.

The overall number of features was reduced from 23 to 18; they are listed in Tables 10 and 11. Among the remaining numeric features, we observe in the heatmap in Figure 11 that *numVotes* positively correlates to *totalImages* (0.61, moderate to high correlation), to *numRegions* (0.56, moderate correlation), and to *totalCredits* (0.51, low-moderate correlation).

2 CLUSTERING

2.1 Centroid-based Clustering

In analyzing the dataset using centroid-based methods, we experimented with three different algorithms: K-Means, Bisecting K-Means, and X-Means. For each experiment, we followed the same steps:

1. Choosing the attributes: we experimented by using both all numeric features (vide Table 10), log-transformed if necessary as detailed in Subsection 1.3, and then by only using a selection of them. In fact, in light of our findings in Subsection 1.5, we dropped *totalImages*, *numRegions* and *totalCredits*, since they are all moderately correlated with *numVotes*. Using this selection of features has yielded better performances.

2. Scaling the dataset: we used min-max normalization, since it lead to better Silhouette scores in comparison to StandardScaler and RobustScaler.

3. Selecting the best value for k (except for X-Means): we used both the Elbow method and the Silhouette method as heuristics to determine the appropriate number of clusters. To do so, we iteratively run the algorithm for

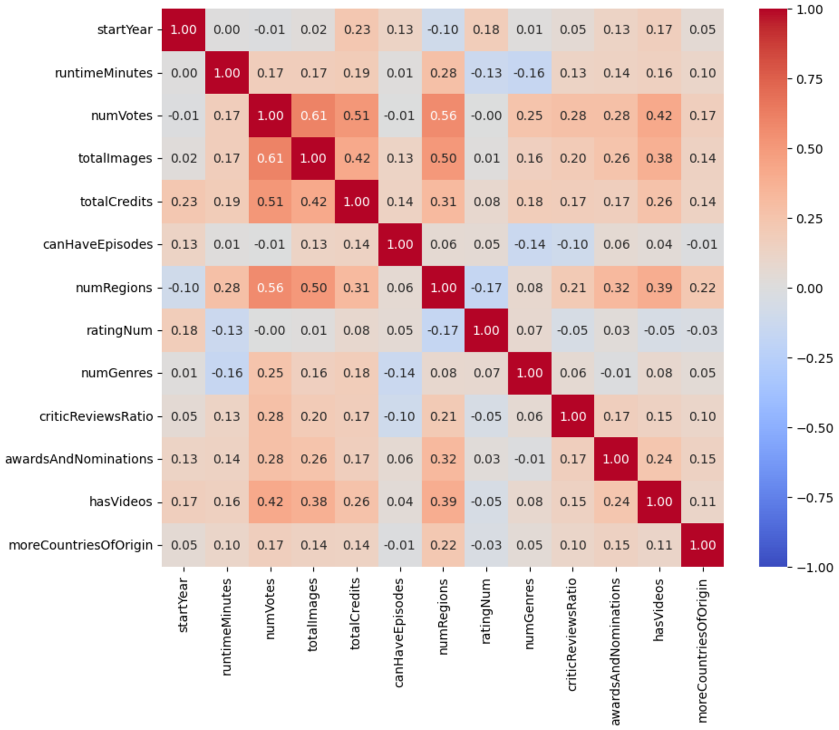

Figure 11: Correlation heatmap after attribute removal.

k in the range [2,50], plotting the SSE and Silhouette score for each *k* and comparing the two curves to find the best trade-off. An example is reported in Figure 12.

4. Fitting the algorithm and evaluating the resulting clustering in terms of cohesion and separation by calculating the SSE, Silhouette score, and correlation between the distance matrix of the dataset and the ideal similarity matrix.

5. Visual cluster exploration: cluster visualization through PCA, parallel coordinates plot of the centroids, and silhouette plots for each cluster.

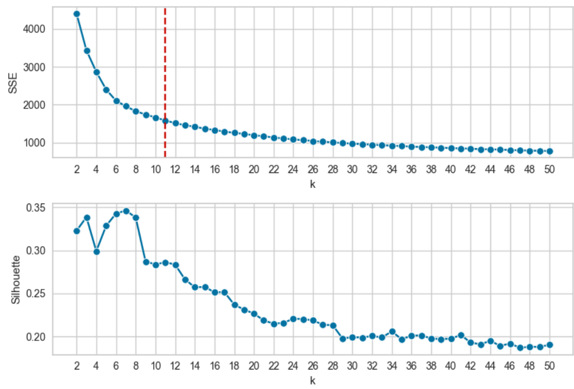

Figure 12: SSE and Silhouette score for choosing *k* (K-Means, all feats.)

2.1.1 *K-Means*

For the first experiment using all numeric features, we select k=8 as a trade-off between SSE and Silhouette score, by analyzing the plots in Figure 12. The resulting SSE and Silhouette score are 1843.054 and 0.333, respectively, while the correlation is -0.616.

The clusters' composition is reported in Table 12: in terms of population, we can distinguish Cluster 2 as the largest cluster, and then three groups, with Clusters 5, 6 and 7 having slightly less than 1000 records, Clusters 0 and 3 having less than 1500, and Clusters 1 and 4 having between 3000 and 3500 records. The first thing that emerges from the PCA visualization and the parallel coordinates graph is that the clusters separate points mostly based on *numGenres* (which only has three discrete values, 1, 2, or 3). *criticReviewsRatio* also leads to interesting centroid separation. Cluster 7 is characterized by a significantly lower value for *startYear*, meaning that it clusters older titles; its centroid is also separated by a lower value for *totalCredits* and *logRuntime*, though not as dramatically.

For the second experiment, using the selection of features, we also select k=8. All evaluation metrics improve: the resulting SSE is 989.711, the

Feature	Description	Type
runtimeMinutes	Primary runtime of the title, in minutes.	Continuous. Ratio-scaled.
startYear	Represents the release year of a title. In the case of TV Series, it is the series' start year.	Discrete. Interval.
numVotes	Number of votes the title has received.	Discrete. Ratio-scaled. Log-transformed.
numRegions	The regions number for this version of the title.	Discrete. Ratio-scaled. Log-transformed.
totalImages	Total number of images for the title within the IMDb title page.	Discrete. Ratio-scaled. Log-transformed.
totalCredits	Total number of credits for the title.	Discrete. Ratio-scaled. Log-transformed.
criticReviewsRatio	The proportion of reviews from critics on the total number of reviews of a record. For unreviewed records, it is set to 0.	Continuous. Ratio-scaled.
awardsAndNominations	Total number of awards and nominations.	Discrete. Ratio-scaled. Log-transformed.
numGenres	The number of genres listed for the variable *genres*.	Discrete. Ratio-scaled.

Table 10: Final numeric attributes.

Feature	Description	Type
originalTitle	Original title, in the original language.	Categorical; mostly unique classes (i.e. names).
titleType	The type/format of the title (e.g. movie, short, tvseries, tvepisode, video, etc.).	Categorical, 10 classes.
countryOfOrigin	The country where the title was primarily produced.	Categorical, 153 classes. Some titles can belong to more than 1 class.
genres	The genre(s) associated with the title (e.g., drama, comedy, action).	Categorical, 29 classes. Some titles can belong to more than 1 class.
rating	IMDB title rating class.	Ordinal, 10 values.
ratingNum	Mapping of *rating* to the integers [1,10].	Ordinal, 10 values.
canHaveEpisodes	Whether or not the title can have episodes.	Asymmetric binary.
hasVideos	Whether or not the title IMDb title page has at least one video.	Asymmetric binary.
moreCountriesOfOrigin	Whether or not the title was produced in more than one country.	Binary.

Table 11: Final categorical, ordinal, and binary attributes.

Silhouette score is 0.444, and the correlation is -0.656. Figure 13 shows that our clustering's SSE is significantly lower than the SSE found in the clustering of 500 randomly generated datasets, which suggests that the clusters found by the algorithm in our dataset are meaningful and not a product of chance.

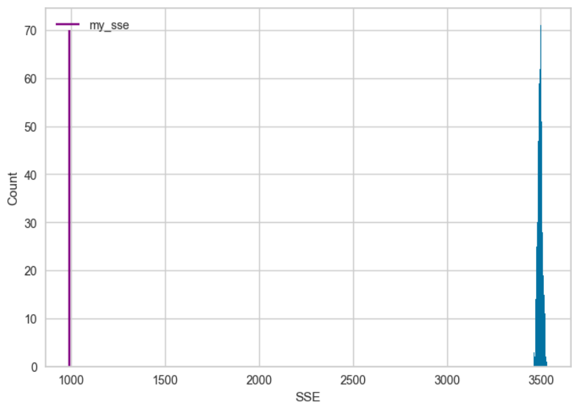

Figure 13: Statistical SSE evaluation for K-Means (k=8, selected feats.)

The clusters' composition, reported in Table 13, shows once again a significantly larger cluster, Cluster 6, with 5115 records, followed by Cluster 7 and 0, having over 3000 records each. The remaining clusters have between 853 and 1076 records. Looking at the silhouette plots in Figure 14, we can say that Clusters 6, 7, and 0 are particularly well-separated, with a significant portion of their data points extending beyond the average Silhouette score, whereas Clusters 1, 2 and 3 seem to be less well-defined.

Again, the PCA visualization and the parallel coordinates graph (Figure 15 and 16) show that the most relevant feature for clusters' separation is *numGenres*, followed by *criticReviewsRatio* (espe-

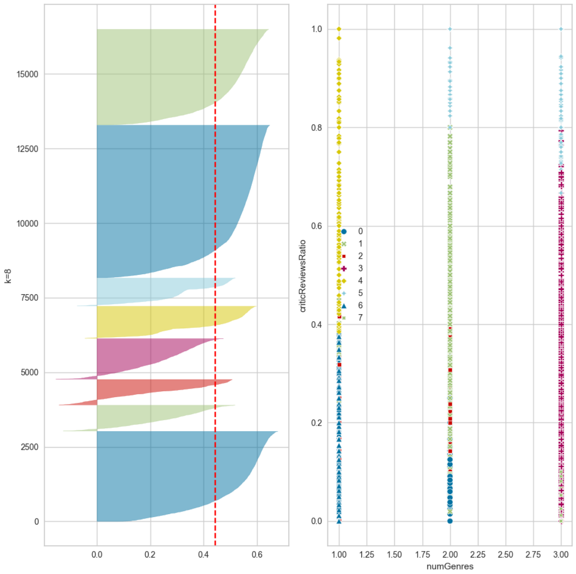

Figure 14: Silhouette visualizer and Scatterplot for the most significant feats(K-Means, k=8, selected feats.)

cially for Clusters 5 and 4). This time, Cluster 2's centroid is particularly separated due to its low value for *startYear*, and, again, is also characterized by a lower value for *logRuntime*.

Cluster	# of records
0	1273
1	3105
2	4511
3	1453
4	3477
5	839
6	909
7	864

Table 12: Clusters for K-Means (k=8, all features).

Cluster	# of records
0	3020
1	865
2	853
3	1360
4	1076
5	935
6	5115
7	3207

Table 13: Clusters for K-Means (k=8, selected feats).

2.1.2 *Bisecting K-Means*

For the first experiment, using all numeric features, we select k=6, even though the elbow for the SSE is k=12, because after that peak the Silhouette score drops significantly. The evaluation metrics are overall disappointing: the SSE is 2142.045, the Silhouette score is 0.337, and the correlation is -0.634.

The cluster's composition is summarized in Table 14: Cluster 3 is the most populated cluster with over 5000 records, followed by Cluster 0 and 4 having both more than 3600, whereas the remaining clusters have between 1197 and 1435 titles. Similarly to before, the clusters' centroids

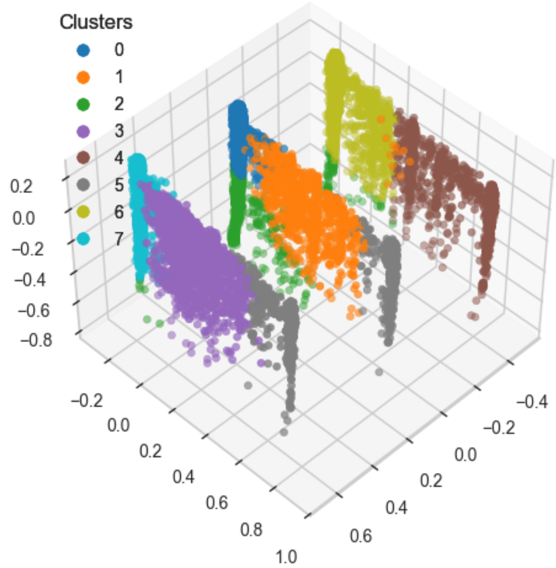

Figure 15: PCA visualization for K-Means (k=8, selected feats.)

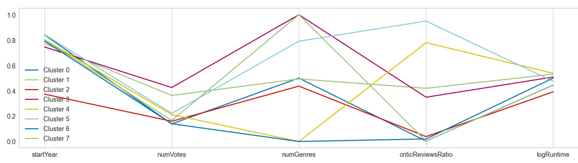

Figure 16: Centroids' visualization for K-Means (k=8, selected feats.)

are mainly separated w.r.t. *numGenres* and *criticReviewsRatio*; Cluster 2 distinguishes itself with particularly high values for *numVotes*, *totalImages*, *totalCredits* and *numRegions*.

The experiment using fewer features, for which we selected k=6, yields better results, with a SSE equal to 1245.326, a Silhouette score equal to 0.436, and the best correlation among all experiments (-0.672). We have two larger clusters, 0 and 5, while the smallest has 995 features (Table 15). Once again, the centroids separate mainly based on *numGenres* and *criticReviewsRatio*; Cluster 3 has a particularly high value for *numVotes* w.r.t. the others.

Cluster	# of records
0	3616
1	1435
2	1318
3	5116
4	1197
5	3749

Table 14: Clusters for Bisecting K-Means (k=6, all features).

Cluster	# of records
0	5225
1	1088
2	995
3	1447
4	3654
5	4022

Table 15: Clusters for Bisecting K-Means (k=6, selected feats).

2.1.3 X-Means

Since using fewer features has yielded better results thus far, we employed the selected subset of numerical features. In the first experiment, we did not set a limit to the execution of the algorithm; the result was a clustering with 20 clusters, which of course led to a lower SSE (534.554), but a less satisfactory Silhouette score (0.313) and the lowest correlation observed among the experiments (-0.477). Since 20 clusters are difficult to analyze, in the following experiments we set k=8 and k=6. We report the results for k=8, which has yielded overall satisfactory results in comparison to most of the experiments, with a SSE equal to 1032.925, a Silhouette score of 0.436, and a correlation of -0.652. The cluster's composition (Table 16) echoes the ones seen for the previous algorithms, with one larger cluster, a couple of medium-large clusters, and four clusters ranging from 681 to 957 records. The clusters' centroids are, once again, mostly separated based on *numGenres* and *criticReviewsRatio*.

Cluster	# of records
0	930
1	957
2	3199
3	1356
4	3676
5	4852
6	681
7	780

Table 16: Cluster sizes for X-Means with k=8, selected feats.

2.1.4 Discussion and Further Evaluation of the Best Centroid-based Clustering

The performances for the different experiments are summarized in Table 17. The overall best results were achieved using K-Means with k=8, using a selection of features (Experiment 2): it resulted in the best SSE and Silhouette Score, as well as the second highest negative correlation.

Exp.	Algorithm	Feats	SSE	Silh.	Corr.
1	K-M (k=8)	All	1843,054	0,333	-0,616
2	**K-M (k=8)**	**Selected**	**989,711**	**0,444**	**-0,656**
3	B.K-M (k=6)	All	2142,045	0,337	-0,634
4	B.K-M (k=6)	Selected	1245,326	0,436	-0,672
5	X-M (k=20)	Selected	534,554	0,313	-0,477
6	X-M (k=8)	Selected	1032,925	0,436	-0,652

Table 17: Centroid-based clustering analysis results.

In order to evaluate the results, we first visualized the distribution of the clusters against known categorical features (those in Table 11, except for *originalTitle* because of its uninformativeness, and *rating* because it is redundant w.r.t. *ratingNum*) by plotting the clusters with a stratified bar chart against each feature. The distribution doesn't show pure clusters against any of them, but there are some small insights:

- *titleType*: Clusters 0, 2 and 7 show a relevant proportion of shorts. Cluster 0 and especially Cluster 7 additionally have a relevant proportion of tvEpisodes. Clusters 3 and 5 also show a relevant portion of tvEpisodes, but not of shorts. Clusters 4 and 1 are made up by more than 50% movies; their centroids indeed have the highest values for *logRuntime*.

- *ratingNum*: all clusters follow a normal distribution w.r.t. to the rating.

- *canHaveEpisodes*: most of the True records are in the biggest cluster, Cluster 6. For every cluster, they represent a small proportion (especially for 2 and 5).

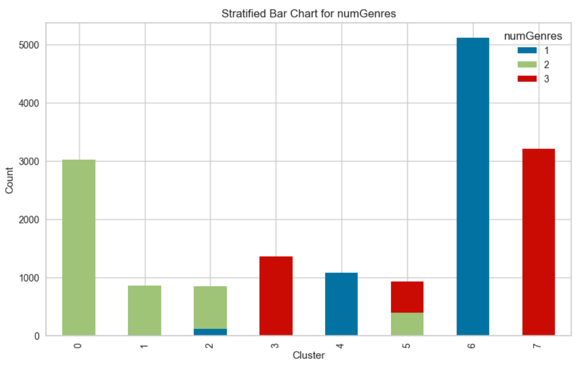

Figure 17: Stratified barchart for *numGenres*.

- *moreCountriesOfOrigin*: not very informative, True represent a small portion for every cluster, but this is especially true for Cluster 2, which is almost entirely False.

- *awardsAndNominations*: Cluster 2 is characterized by being almost entirely made of records with no awards nor nominations.

- *hasVideos*: Cluster 2 is entirely False, whereas Cluster 3 has the most number of True out of any other cluster, showing a significantly higher proportion.

A clear characterization emerges for Cluster 2: it mostly contains shorts (its centroid has the lowest value for *logRuntime*), and it has very little to no True values for every boolean variable. Since it's greatly made up of shorts, then movies and tvEpisodes, we can expect its very low number of True for *canHaveEpisodes*. Its titles are produced in just one country, they have mostly no awards nor nominations (we could expect movies and TV series to gain more accolades) and also almost no video content on their page (again, this could reasonably be a feature for blockbusters and mainstream TV series).

Subsequently, we investigated the clusters' composition w.r.t. numeric variables, first using a pair plot to gain a global view (which confirmed that the clusters are mainly separated on the basis of *numGenres*) and then using appropriate plots to investigate single variables. The stratified barchart in Figure 17 shows how most clusters are pure w.r.t. *numGenres*: 0 and 1 only have 2 genres, 3 and 7 only have 3 genres, while 4 and 6 only have 1 genre. There remains Cluster 2, which mostly has 2 genres with a small proportion of 1 genre, and Cluster 5, with around 60% 3 genres and 40% 2 genres.

The box plot in Figure 18 shows the cluster's distribution w.r.t *criticReviewsRatio*:

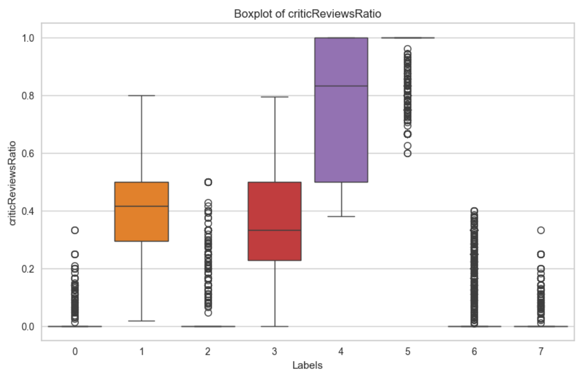

Figure 18: Box plots for *criticReviewsRatio*.

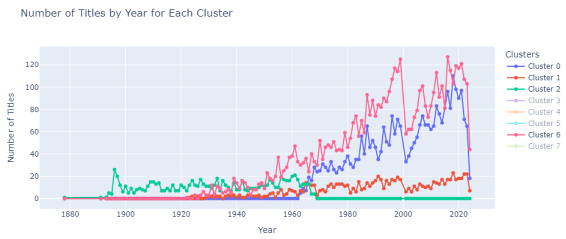

Figure 19: Titles by startYear for relevant clusters.

- Cluster 5 is concentrated at 1.0: the reviews are mostly by critics.

- Cluster 4 also has a high ratio (median at around 82%, most of the records go from 50% to 100%).

- Clusters 1 and 3 show variability, but they are a mostly balanced mix of critic and user reviews. They could represent mainstream content.

- Clusters 0, 2, 6 and 7 are skewed towards 0: their reviews are mostly made by the public. They could represent niche or user driven content (indie or fan-made media).

The cluster's line-graphs for *startYear* (Figure 19) confirm that Cluster 2 is characterized by only grouping titles released between 1880 and 1968. On the other hand, Cluster 0 groups newer titles, released since 1963. Clusters 1 and 6 contain no titles released before the 1920s.

The box plots for *runtimeMinutes* give the following insights:

- All IQRs span from around 15 to 100 minutes.

- Cluster 2 has the lowest median runtime (which is coherent with what we've already discussed), followed by Clusters 0, 5 and 7, which have a median at around 40 minutes – they did in fact contain a large portion of shorts and TV episodes. Cluster 3, with a median of 75 minutes, has been characterized as probably containing mainstream content (balanced *criticReviewsRatio*,, highest number oh *hasVideos*): it has a mostly balanced mix of TV episodes and movies.

- Clusters 1 and 4 have a small IQR and a median around 90 minutes, which relates to their records being mostly movies.
- Cluster 6 has outliers that go to over 1000 minutes; it seems that the largest group is also the most heterogeneous. It can be linked to its significant portion of TV series, the runtime of which has been variably identified as the total runtime of the series, or the runtime per episode.

With regards to *numVotes*, the box plots for the different clusters don't add much information to the centroids' visualization: Clusters 0 and 6 have lower medians, but many upper outliers; Cluster 7, followed by Cluster 3, have the highest medians, but their large whiskers indicate that the data is very spread out.

3 CLASSIFICATION

4 REGRESSION

5 PATTERN MINING